

ABSORBING INTEGRATION: EMPIRICAL EVIDENCE ON THE MEDIATING ROLE OF ABSORPTIVE CAPACITY BETWEEN FUNCTIONAL-/CROSS-FUNCTIONAL INTEGRATION AND INNOVATION PERFORMANCE

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Integration of organisational units has been extensively researched in various streams of management and organization sciences. It is very important whenever knowledge differences have to be overcome due to functional departmentalisation and the ensuing knowledge specialisation. However, extant literature does not yet appreciate the mediating role of absorptive capacity (AC) in this context. We argue that departments use integration mechanisms in order to develop and maintain such an organisational capability to absorb knowledge from other departments, so that integration can succeed to increase innovation performance. Our unique dataset of Italian manufacturing firms from various industries allows us to study this in the context of the integration of research and development (R&D) and marketing and sales (M&S) departments. Thereby, we provide empirical evidence on the mediating role of AC. We find evidence that R&D departments build AC via formal cross-functional integration, while M&S departments do so through informal integration. Moreover, we provide evidence of AC's mediating role for the relationship between cross-functional integration mechanisms and innovation performance. Our findings also reveal significant differences between R&D and M&S functions in terms of effect sizes and

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significance levels. AC of R&D departments has a significant and substantial effect on innovation performance and thus effectively acts as a mediating variable, while in case of M&S departments we observe a significant direct effect between formal cross-functional integration and innovation performance without any mediation by AC.

Keywords: Innovation performance; cross-functional integration; Absorptive capacity; structural equation modeling.

Introduction

Specialisation offers a substantial efficiency advantage, but also creates silos of specialised knowledge (Lessard and Zaheer, 1996). In order to remain innovative, it is fundamental for firms to successfully integrate these departments without compromising the benefits of specialisation (Burns and Stalker, 1961; Galbraith, 1974; Iansiti and Clark, 1994; Lawrence and Lorsch, 1967; Lee and Kapoor, 2017). Integration is “the process of achieving unity of effort among the various sub-systems [functional departments] in the accomplishment of the organization’s tasks” (Lawrence and Lorsch, 1967, p. 4). However, knowledge specialisation can lead to conflicting or incompatible thought worlds (Dougherty, 1992), impede effective communication (Bechky, 2003), and raise several barriers for comprehension and cooperation (Carlile, 2004). Therefore, integration mechanisms have to either transcend or traverse these knowledge differences (Majchrzak *et al.*, 2012).

For the scope of our study, we differentiate between corporate functions and departments. Corporate functions are intra-firm systems that are differentiated relative to the requirements of the particular environment with which they deal (Lawrence and Lorsch, 1967). Examples for such corporate functions are research and development (R&D) and marketing and sales (M&S). Departments are sub-systems within such functions (Clark and Wheelwright, 1992). For example, the “lighting technology R&D” and the “driver assistance systems and automated driving R&D” departments of an automobile manufacturer both are departments belonging to the R&D function. Knowledge differences obviously exist between departments of different corporate functions. However, they also exist between departments of the *same* corporate function. In fact, many studies show that intra- and cross-functional integration increase innovation performance, i.e., the degree to which innovation objectives are met (Atuahene-Gima *et al.*, 2005; Foss *et al.*, 2011).

However, substantial divergence persists in findings regarding the relation between cross-functional integration and new product success, and, ultimately innovation performance (Rubera *et al.*, 2012; Troy *et al.*, 2008). Some argue that higher functional integration is always more beneficial for the firm (e.g., Song and Parry, 1997). Others, by contrast, suggest that the optimal degree of integration is

not necessarily its maximum (e.g., Griffin and Hauser, 1996). Several studies have even found negative outcomes of cross-functional integration (Bommer *et al.*, 2002; Hansen, 2009; Hoegl *et al.*, 2004). This divergence might be explained by (a) the different processes or mechanisms used to operationalise integration, as well as (b) the different nature of the organisational units to be integrated.

Indeed, regarding (a), previous research studied several different integration mechanisms, including not only functional and cross-functional but also formal and informal ones (Griffin and Hauser, 1996; Leenders and Wierenga, 2002). Examples of *formal* mechanisms are job rotation, regular meetings, and cross-functional project teams. Examples of *informal* mechanisms are ad hoc lateral communication, bypassing official channels, or the maintenance of open communication channels in the day-to-day operations with other departments. As a result, previous research found different effects for different integration mechanisms. Moenaert *et al.* (1994) report a positive impact of formalisation on communication flows, while Maltz *et al.* (2001) could not find support for their hypothesis of a positive impact of informal integration in form of socialisation opportunities on the conceptual use of marketing information by R&D managers. Other studies are agnostic towards the actual processes and operationalise integration by the level of interaction (Lamore *et al.*, 2013; Li and Calantone, 1998).

Regarding (b), R&D and M&S are two types of corporate functions that seem to offer particularly high returns to integration. This is apparently due to their high complementarity. Griffin and Hauser (1996) state that their “responsibilities in the new product development are neither independent nor static; they cannot be analyzed separately” (p. 192). Consequently, a large number of studies focus on both R&D and M&S, many of which demonstrate that their integration has a positive effect on innovation performance (Galbraith, 1974; Griffin and Hauser, 1996; Gupta *et al.*, 1986; Jansen *et al.*, 2009; Lawrence and Lorsch, 1967; Ruekert and Walker, 1987). Similarly, findings from the literature on market orientation underline the crucial role that marketing plays in increasing performance (Jaworski and Kohli, 1993; Menon and Varadarajan, 1992; Verhoef and Leeflang, 2009). Just as earlier studies mostly focused on R&D-marketing integration, more recent ones likewise suggest that R&D-sales integration is highly relevant for new product development (Homburg *et al.*, 2017). Research also shows the importance of integration of complementary units with similar functions, but covering different product markets, such as in multiproduct firms (Lee and Kapoor, 2017). However, what has been neglected so far is the role of organisational capabilities, such as that the capability to absorb knowledge in case of potentially asymmetric task interdependencies between such complementary departments.

Hence, focusing on formal and informal integration mechanisms in R&D and M&S departments, we address both of these explanations for persistently

divergent findings regarding the benefits of functional integration. We explain the effects of different integration mechanisms on innovation performance arguing that organisational units of different functional natures can use these mechanisms to different degrees to build capabilities required for integration. Given the fundamental importance of knowledge for innovation performance, we suggest that the capability to absorb knowledge from other units is particularly important for innovation performance.

Absorptive Capacity (AC) is a firm's ability to recognise the value of external knowledge, assimilate it, and exploit it to commercial ends (Cohen and Levinthal, 1990). We argue that this ability not only exists at a firm's external boundaries where such a capability can foster for example technology transfer (Bauer *et al.*, 2018). It also exists beyond the firm-level at lower organisational levels as well, such as the department-level. Without a sufficiently high degree of AC, the costs of coordination associated with integration can have negative effects on innovation performance that are not outweighed by its benefits, like overspending, project delay, and project termination (Cuijpers *et al.*, 2011). However, while research intensely investigated AC (e.g., Adams *et al.*, 2016; Todorova and Durisin, 2007; Volberda *et al.*, 2010; Zahra and George, 2002), AC literature is limited to analyses at the firm or business unit level. Only recently, research started to investigate the micro-foundations of AC (Distel, 2018). However, the explicit consideration of the nature of the absorbed knowledge as well as the analysis of AC at levels below the business unit are important and yet poorly explored subjects (Volberda *et al.*, 2010). This raises important research questions: *Can the capability of departments to absorb knowledge from other departments improve innovation performance? Which integration mechanisms do actually contribute to the development of such an ability? Is the relevance of AC symmetric or does it follow potential asymmetries in task interdependence?* In this study, we aim to answer these research questions and thus make several important contributions to research on AC as well as to research on functional and cross-functional integration.

Theoretical Framework

Integration mechanisms and innovation performance

In line with previous research, we consider four types of integration mechanisms. First, we distinguish between formal and informal ones (Bauer *et al.*, 2018; Burgers *et al.*, 2009; Moenaert *et al.*, 1994) and, for each, we further distinguish between intra- and cross-functional integration mechanisms (Hoegl *et al.*, 2004; Olson *et al.*, 2001). These integration mechanisms can serve either to coordinate tasks of interdependent functional departments or to transfer knowledge between

them. However, also in case of mere coordination, the departments need to exchange information. Hence, the sending and receiving of information and knowledge is at the core of all integration mechanisms. Several studies have shown the positive effects functional integration can have. The literature on team and group behaviour investigates the direct impact of different kinds of integrating processes on performance (Ancona and Caldwell, 1992), such as boundary spanning behaviour (Allen and Cohen, 1969; Tushman, 1977; Zhao and Anand, 2013). Further empirical findings have corroborated the direct effect of integration on innovation in form of corporate entrepreneurship (Burgers and Covin, 2016; Foss *et al.*, 2011).

However, the richness of integration mechanisms, such as their capacity to transfer a high variety and amount of information, comes at the cost of higher complexity (Tushman and Nadler, 1978). Moreover, Sherman and Keller (2011) have shown that managers often misconceive the degree to which their unit's tasks and the tasks of other functional units are interdependent and may, therefore, choose wrong degrees of integration. Ultimately, this can lead to lower performance as is explained in several studies reporting negative effects of integration mechanisms (Troy *et al.*, 2008). The successful exchange of information and knowledge is necessary to carry out the innovation related tasks that drive innovation performance. For this exchange to be successful, receiving departments need to have the capability to absorb the incoming information and knowledge.

Absorptive capacities of functionally specialized departments

Cohen and Levinthal (1990) first conceptualised AC as the ability of firms to *recognise the value* of external knowledge, *assimilate* it, and *exploit* it to commercial ends. Most of the extant research on AC treats prior related knowledge as the most fundamental antecedent of AC (Todorova and Durisin, 2007; Zahra and George, 2002). This is in part because most studies investigate AC at the firm level, for which it had been initially conceptualised. At this level, organisational knowledge is most aggregate and hence analysed in a correspondingly general manner. Despite this high level of analysis, extant studies on AC almost always implicitly focus on technological knowledge.

In fact, Cohen and Levinthal (1990) deemed R&D and manufacturing operations the major sources of AC because these provide “prior related knowledge”, which refers to such knowledge present in an organisation prior to absorption and related to the target knowledge. However, while this focus is justified at the organisational boundary, where firms try to absorb external technological knowledge relevant for new innovations, it falls short in explaining the relevance of AC within the firm, where M&S needs feedback on what is technologically

feasible in order to make better market predictions and R&D needs updates on how the market conditions and customer expectations have changed. Yet, no study so far has conceptualised AC as a phenomenon on the level of departments of different functional specialisations, limiting the scope to one functional setting or to higher levels, e.g., the transfer of more or less sticky practices among operational units (Szulanski, 1996) or across subsidiaries of multinational corporations (Gupta and Govindarajan, 1991, 2000).

Moreover, different functions have different goals and incentives and are in different degrees senders and receivers of knowledge (Gupta *et al.*, 1986; Song *et al.*, 2001; Troy *et al.*, 2008). The departments are also embedded in completely different communities of practice (Brown and Duguid, 2001) and they may use different narratives and stories to interpret their environment and the knowledge available to them (Beverland *et al.*, 2016; Boland and Tenkasi, 1995; Dougherty, 1992). Hence, barriers¹ to knowledge integration can arise from different vocabulary (syntactic barriers); different meanings (semantic); or different interests (pragmatic) (Carlile, 2002, 2004).

Therefore, the challenges at departmental boundaries *within* corporate functions like the R&D function (e.g., interactions and knowledge integration between two R&D departments) are fundamentally different from the challenges at inter-departmental boundaries *across* corporate functions (Brettel *et al.*, 2011; Gupta *et al.*, 1986), like the boundary between an R&D and an M&S department. As a consequence, the involved unit's capability to absorb knowledge needs to be conceptualised as two distinct abilities in order to explain why the same integration mechanisms may have different effects in different contexts. We thus assert that

Functional AC (FAC) and Cross-Functional AC (CFAC) refer to the ability of a functionally specialised department to value, assimilate and apply the functionally specialised knowledge available from other departments of the SAME or a DIFFERENT corporate function, respectively, in order to fulfil the department's tasks.

Antecedents of functional and cross-functional ACs

In line with Griffin and Hauser (1996), who suggest functions need to integrate in case of environmental uncertainty, Rubera *et al.* (2012) find that integration is particularly useful in case of exploration of new competences. However, AC is

¹While Carlile uses the term “boundaries,” we use “barriers” in order to clearly distinguish these mostly cognitive and communicative barriers to integration from organisational/structural boundaries.

assumed to depend principally on prior related knowledge and absorbing units have qua definition little *prior* knowledge when they explore new fields. Hence, AC is lowest when it is needed most. The degree to which different integration mechanisms are used affects the degree of specific capabilities related therewith. Hence, we argue that the ability to absorb new knowledge not only depends on what the absorbing unit already knows on a topic (*prior related factual knowledge*), which may be helpful for resourceful sensemaking (Beverland *et al.*, 2016), but it also depends on *prior related procedural knowledge*. This can be knowledge regarding which useful new knowledge resides in which other departments, how to get access to this new knowledge, or how to interact with the source department.

Moreover, since AC depends on prior *related* knowledge it is not generally high or low for a specific organisational unit, but always specific to a particular knowledge domain. That means that, an R&D department can exhibit high AC regarding the knowledge of a fellow R&D department from a different business unit and at the same time exhibit low AC regarding the knowledge of an M&S department, possibly due to potential tensions between the different personalities of marketers and engineers (Shaw and Shaw, 1998). Cohen and Levinthal (1990) focused on R&D and mostly ignored the influence of M&S, leading to AC being studied in relation to technological knowledge only, at least implicitly. While several studies find that innovation and performance are positively affected by AC (Rothaermel and Alexandre, 2009), AC literature lacks empirical studies that explicitly consider the type of knowledge being absorbed, such as technology and market knowledge. Therefore, we theorise and operationalise the concept of AC at the level of functional departments in the form of two distinct abilities — functional and cross-functional AC — and reveal their mediating role in the relationship between integration mechanisms and innovation performance.

Hypotheses

Effects of intra-functional integration mechanisms on departmental ACs

The important ability to integrate knowledge is essentially based on communication capabilities (Daft and Lengel, 1986). Intra-functional communication is regarded as an important element in the R&D process (Allen and Cohen, 1969; Szulanski, 1996) and particularly relevant in the intermediate phase of the innovation process (Tushman, 1977). Organisational units at any level have to develop and improve these capabilities, and there is substantial evidence that organisations learn through experience (Ingram and Baum, 1997); through the repetition of a specific behaviour like integration mechanisms. Both formal and informal integration mechanisms are notable forms of organisational memory (Moorman and

Miner, 1998; Walsh and Ungson, 1991). Consequently, they not only are subject to depreciation but also can be forgotten if not applied continuously. All intra-functional integration mechanisms have in common that they require the absorption of knowledge from functional peers; that is, from departments that are specialised within the same corporate function, such as two R&D departments of different product divisions. The more they interact with each other and exchange knowledge, be it by means of formal or informal integration mechanisms, the more sophisticated they should become in the use of these mechanisms and, hence, the better they will be able to absorb knowledge from their peer departments. Therefore, it has to be assumed that the extensive use of intra-functional integration mechanisms in general entails a higher departmental FAC.

Hypothesis 1: *The more extensively a department uses either (a) INFORMAL or (b) FORMAL intra-functional integration mechanisms, the higher its FAC.*

Effects of cross-functional integration mechanisms on departmental ACs

It has been shown that R&D managers use market information that is perceived as being of high quality both instrumentally, for immediate decisions; and conceptually, to adapt their view of the market environment (Maltz *et al.*, 2001). For example, in the case of a disagreement in the R&D department regarding the potential of a particular technical improvement or development path, integration mechanisms might help to obtain a fast expert opinion on the market potential of technological developments. Informal cross-functional integration mechanisms have the potential to provide departments with feedback on ideas previously out of search scope that might help to reconcile conflicting interests because of their potential to be used ad hoc much more quickly and more spontaneously. Hence, informal cross-functional integration mechanisms also help receiving departments to better understand the value of incoming knowledge from their own functional area, thus supporting the integration with other departments belonging to the same corporate function.

Hypothesis 2: *The more extensively a department uses informal cross-functional integration mechanisms, the higher its FAC.*

In different departments, systematically different personalities exist, such as marketers and engineers (Shaw and Shaw, 1998); differences between them must be considered and understood in order to avoid misunderstandings and conflicts. Moreover, in cross-functional integration, different departmental thought worlds face each other (Dougherty, 1992), so departments must make a greater effort to

develop knowledge on how to deal with different types of departments to a greater extent than in intra-functional integration.

In turn, the way in which AC builds through the use of *formal* integration mechanisms differs from that of informal integration mechanisms. Formal mechanisms can enhance, above all, the information flow and knowledge exchange in predetermined directions (Moenaert *et al.*, 1994). An essential part of cross-functional integration is knowledge exchange between functional units that are task interdependent: those units that depend on each other in their task accomplishment (Fry and Slocum, 1984). However, task interdependence is often asymmetric (De Jong *et al.*, 2007), so that the required knowledge flows can be asymmetric, too. This is particularly pronounced in the case of the M&S-R&D interface. Understanding feedback from M&S departments is vital in many stages and types of innovation projects (Wang, 1997). In fact, one important conclusion according to the marketing-R&D interface literature is that “(. . .) it is through the marketing-R&D interface that market knowledge is transferred, to and integrated with technological knowledge” (Li and Calantone, 1998, p. 15). What is more, formal integration mechanisms are an important means to make market knowledge available to the R&D function (Ernst *et al.*, 2010; Li and Calantone, 1998; Song *et al.*, 1998). The direction of knowledge from M&S to R&D is most obvious in the market-pull approach to new product development (Rothwell, 1994). With knowledge flowing primarily from M&S to R&D and much less so vice-versa, strong CFAC is more relevant for R&D than for M&S departments. The very use of integration mechanisms can lead to learning by doing (Ingram and Baum, 1997), thereby increasing departmental AC. Therefore, we expect R&D departments to be more often the absorbing part when using integration mechanisms, thus building CFAC; in contrast, M&S departments use these mechanisms mostly to *provide* knowledge, not build CFAC.

Hypothesis 3(a): *The more formal cross-functional integration mechanisms are used by R&D departments, the higher is their CFAC, while more extensive use of these mechanisms by M&S departments has no impact on CFAC.*

Informal integration mechanisms can positively impact cross-functional integration to the same degree as formal ones (Pinto and Pinto, 1990; Zahra and Nielsen, 2002). In particular, M&S departments that do not improve CFAC through formal cross-functional integration mechanisms have the chance to get more insight into the thought-worlds and practices of R&D departments when they engage informally with them. However, there are at least three reasons why R&D departments might suffer in their development of CFAC when they engage in informal cross-functional integration mechanisms.

The first reason is that the informal nature of a communication channel might provide opportunities and therefore tempt individual employees to advance their personal agendas or even induce them to send more information than through formal communication channels (Johnson *et al.*, 1994). In these cases, informal communication channels can produce an information overload for departments that already receive information through formal mechanisms. Such information is, furthermore, potentially contrasting and confusing as it is conveyed informally in an unstructured form. Altogether, the overload via unstructured information ties all attention and resources of the receiving department, thereby preventing reflection of the knowledge absorption process itself and, ultimately, leading to a collapse of sensemaking (Weick, 1993).

Secondly, a negative impact of informal cross-functional integration mechanisms on CFAC might be the result of the different types of knowledge involved. Technological knowledge is very much path dependent (Dosi, 1982), which explains the focus on prior related knowledge in AC literature as it is concerned principally with technological knowledge. Market demands, however, might change radically over time, thus more frequently creating disruptions (Avagyan *et al.*, 2016; Christensen, 2003). Evidence suggests that companies fail to adapt to radically changing environments, despite having the technological capabilities to do so, because new market knowledge is rejected by key actors or units (Tripsas, 2009). Formal cross-functional integration mechanisms can challenge such outdated market knowledge outright, although not without conflict. Informal communication, however, offers more possibilities to avoid conflict and to communicate only with less challenging sources that, rather, confirm existing convictions. This is consistent with the tendency known as confirmation bias: the “tendency, when testing an existing belief, to search for evidence which could confirm that belief, rather than for evidence which could disconfirm it.” (Jones and Sugden, 2001, p. 59).

Thirdly, recent empirical evidence suggests that informal communication might disrupt new product development particularly in cases of complex products (Sosa *et al.*, 2015). This occurs when informal communication provides cyclic exchange across a social network, as often happens when teams responsible for the early stages of the innovation process, such as the development stage, receive informal feedback from teams responsible for the later stages, such as, the commercialisation stage. This continuously interrupts the innovation process rather than creating higher market orientation. Consequently, we expect the effect EYJLEJ of informal cross-functional integration mechanisms on CFAC to be positive only for M&S, while we expect it to be negative in the case of R&D departments.

Hypothesis 3(b): *The more informal cross-functional integration mechanisms are used by R&D departments, the lower their CFAC is, while more extensive use of*

informal cross-functional integration mechanisms by M&S departments positively impacts CFAC.

Effect of departmental ACs on innovation performance

FAC and CFAC should be closely related. FAC can be considered as a kind of fundamental AC of a functionally specialised department. While FAC enables the absorption of knowledge from other departments of the same functional specialisation, this ability impacts the ability to absorb knowledge from other corporate functions, i.e., CFAC. The understanding of the process of absorption in general without the confounding element of different corporate functions can be regarded as the prior related procedural knowledge with respect to how to recombine, transform, and exploit new extra-departmental knowledge internally. This allows then for the more efficient absorption of complementary, cross-functional knowledge. Therefore, we specify:

Hypothesis 4: *The higher the departments' FAC, the higher their CFAC.*

Once the departments have developed FAC, they can valorise potential synergies and complementarities existing between their own and other departments' corporate function by absorbing their knowledge. Empirical studies of firms' sub-unit's AC provide evidence that the recipient's AC is a major determinant of the success or failure of intra-organisational knowledge transfer (van Wijk *et al.*, 2008). This finding should be valid for both FAC and CFAC and should also apply to both technological and market knowledge. For example, Luo *et al.* (2006) find that interdepartmental *cooperative ability* regarding market knowledge among departments — defined in terms of AC — positively affects both customer and financial performance.

However, in context of cross-functional integration, M&S and R&D are not equally likely to be recipients of knowledge. While relations have been found between knowledge flows from R&D to M&S and NPD performance (Moenaert *et al.*, 1994), studies analysing the effect at various stages suggest that the role of R&D-M&S integration mainly consists in the provision of market feedback to the R&D department (Brettel *et al.*, 2011; Drechsler *et al.*, 2013; Olson *et al.*, 2001), for example during the stages of market opportunity analysis, development, and pretesting (Song *et al.*, 1998) or, with the goal to improve market orientation (Jaworski and Kohli, 1993) or customer connection (Moorman and Rust, 1999). We may therefore conclude that the knowledge most important to innovation performance flows from M&S to R&D. R&D departments cannot simply participate in integration mechanisms and benefit from these without being able to

actually absorb the knowledge in question, which might be possible for the M&S departments providing knowledge instead of absorbing it. We hence hypothesise:

Hypothesis 5a: *FAC of R&D and M&S departments shows a positive direct effect on innovation performance.*

Hypothesis 5b: *CFAC of R&D departments shows a positive direct effect on innovation performance, whereas CFAC of M&S departments shows no direct effect on innovation performance.*

Indirect effects of integration mechanisms

The above discussion implies some potential mediation effects. The consequence of the direct effects from informal cross-functional integration mechanisms on FAC, and from FAC on CFAC is that informal cross-functional integration mechanisms indirectly positively affect cross-functional AC in the case of R&D departments. This means that in turn it could counterbalance the negative direct effect that we predict informal cross-functional integration mechanisms could exert on CFAC of R&D departments. Thus, we put forward the following related hypothesis:

Hypothesis 6: *For R&D departments, there is a positive indirect effect from the use of informal cross-functional integration mechanisms via FAC on CFAC.*

Finally, CFAC of R&D departments mediates the effects of two antecedents of innovation performance: the effects of FAC and formal cross-functional integration mechanisms. Cross-functional integration aims at providing the necessary knowledge directly through lateral relations to those who need it in other functions due to task interdependence. Hence, in order to valorise formal cross-functional integration mechanisms, knowledge must be absorbed successfully. As argued above, the mere collaboration without understanding is not sufficient for a receiving unit. In this case, the receiving unit is hypothesised to be the R&D unit, which heavily relies on market information from M&S departments to direct and orient its work towards current and future market demand. As information providers, the M&S departments, in turn, are less dependent on a precise understanding of technological knowledge. Thus, if the data confirm a direct effect of integration mechanisms for M&S departments — as hypothesised above — while, at the same time, negating such an effect for R&D departments (supporting instead the hypothesis of an indirect effect through CFAC), this would clearly support the thesis that the direction of knowledge flow from M&S to R&D is more important than the reverse.

Hypothesis 7a: *FAC of R&D departments shows a positive indirect effect on innovation performance via CFAC.*

Hypothesis 7b: *There is a positive indirect effect from R&D departments' use of formal cross-functional integration mechanisms via CFAC on innovation performance.*

Direct effects of integration mechanisms on innovation performance

Since our conceptualisation of departmental AC is derived from the firm-level AC, it likewise includes not only the search and evaluation of external knowledge, but also its transfer and exploitation. This would reflect successful integration, which is supposed to improve innovation performance. Therefore, we hypothesised that departmental AC mediates all relevant integration mechanisms. In our model, the mediation should capture the entire effects from integration mechanisms on innovation performance. The presence of direct effects would imply that integration mechanisms achieve more than just (cross-) functional integration.

However, given that M&S departments are usually the party providing access, they are less required to build ability for absorption. In the case of R&D-M&S interfaces, where innovation activities are organised according to the market-pull model, it is particularly clear that there might be a *per se* predominant direction of task dependence and hence knowledge flow (Rothwell, 1994). M&S departments' engagement in integration mechanisms, obviously, can only directly result in higher innovation performance if either (i) the absorbing counterpart knows how to absorb or (ii) the integration mechanism is highly formalised and codified so that no particular prior related procedural knowledge is required. Given that we already expect *informal* cross-functional integration mechanisms to have a negative effect on R&D departments' CFAC, the only direct impact from integration mechanisms on innovation performance can be expected to result from the use of *formal* cross-functional integration mechanisms by M&S departments.

Hypothesis 8: *(a/b/c) None of the studied integration mechanisms directly affects innovation performance, (d) with the exception of M&S department's use of FORMAL cross-functional integration mechanisms, which positively affects innovation performance.*

Methodology

Sampling and data collection

We empirically tested our hypotheses using data from the Italian manufacturing industry. In a first step, we selected all Italian manufacturing firms from the AIDA

database of Italian public and private firms. This database has been used in many previous studies (Colombo *et al.*, 2007) and is considered almost exhaustive, as it includes not only publicly listed companies but also privately held SMEs. It includes 3769 firms with at least 200 employees, which was the minimum threshold we had to set since we could include only firms that had at least two departments of each type in focus: R&D and M&S departments. Several firms also had to be removed from this list because they were no longer active. In a second step, an online pool of potential survey participants was accessed, allowing us to select professionals by firm and department. These restrictions reduced our sample to 541 professionals from either R&D or M&S departments. Professionals who had worked less than one year in their current firm were excluded from our study because we could not assume their responses to be sufficiently reliable, as the process of socialisation usually takes some time. The professionals thus matched were then contacted and asked to complete an online questionnaire. The distribution of experience of survey participants is slightly skewed towards shorter professional experience. This is due to an overrepresentation of younger professionals in the database itself. Since this does not relate to our particular research questions we see no reason to suspect any systematic bias.

The questionnaire was hosted on a dedicated server under the official university domain, and it was password protected to signal careful and confidential use of the participants' data. Moreover, in the contact e-mail all participants were not only assured of the confidentiality of their answers, but incentivised also with a personalised benchmark report. There were two rounds of reminder e-mails. The responses received amounted to 140, of which 126 were completed. Of the 126 complete questionnaires, 51 were filled in by M&S professionals while 75 were completed by employees of R&D departments. Thus, we altogether achieved a response rate of over 23%, which is seen also in similar studies using online surveys (e.g., Verhoef and Leeflang, 2009) and generally a good rate for online surveys of managers, given the decline in online survey response rates since the introduction of the Internet (Sheehan, 2001).

The sectors represented in our final sample include automotive and suppliers, food and beverages, consumer electronics, instruments and industrial machinery.² The sectoral composition of respondents does not differ much from that of non-respondents. We tested for non-response bias using financial data from the AIDA database, since this data was also available for non-respondents. The variables chosen were major performance indicators such as sales, EBIT, and ROE, but also

²The 2-digit ATECO codes of the sectors represented in our sample are the following: 10, 15, 17, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33.

indicators of differences between sectors. This analysis was performed by means of a paired-sample *t*-test on mean differences for each of the selected key variables regarding both the entire sample and the two sub-samples of respondents from R&D and M&S departments. At no point were significant differences found, and we concluded that there is no non-response bias.

In order to avoid any biases related to the sequence of items in a battery, all item batteries used were presented in a random order. Moreover, it was important to avoid the creation of an overly lengthy questionnaire that could have increased the number of response interruptions and, consequently, could have led to a higher number of incomplete responses. The median response time, though, is more informative and was approximately 15 min. Finally, construct validity was assessed in two steps. In a first step, the questionnaire was discussed with senior researchers from both innovation management and marketing. In the second step, a pretest was carried out with several professionals who were afterwards interviewed about the comprehensibility and validity of the constructs. Both researchers and professionals were Italian native speakers with an excellent command of the English language and were also asked to confirm the validity of the translation. However, the questionnaire language could be chosen and changed online by the respondents. Good construct validity can thus be assumed.

Operationalisation

With the exception of the newly established construct of departmental ACs (i.e., FAC and CFAC), all variables have been measured based on previously validated items used in the management literature (see Appendix). The scales for the new concepts FAC and CFAC have been constructed on the basis of items established in the literature measuring AC at the organisational or sub-unit level and adapted slightly to fit the context of the functionally specialised departments chosen: R&D and M&S (Flatten *et al.*, 2011; Hansen and Nohria, 2004; Jansen *et al.*, 2005; Szulanski, 1996).

Innovation performance was assessed by means of a set of items chosen to measure innovation performance, based on instruments that are used in both marketing and management literature (Atuahene-Gima *et al.*, 2005; De Luca and Atuahene-Gima, 2007). It was measured as the degree to which the stated objectives of innovation projects have been reached in the following four dimensions: market share, sales, return on investment, and product performance.

Luo *et al.* (2006, p. 72) conceptualised cross-functional cooperative ability as the “ability to assimilate and deploy market knowledge in lateral interactions among functional areas”, a special case of our more general CFAC. However, this definition refers exclusively to market knowledge, so that it is not suited to

distinguish between FAC and CFAC. As a consequence, the items used by Luo and colleagues do not distinguish between the department's own and other functional knowledge domains. In contrast, the present study focuses on this distinction, aimed at measuring FAC and CFAC explicitly as two distinct concepts. Particular care was taken to select only reflective items from the previous literature.

Furthermore, we made sure that these items were coherent with the theoretic conceptualisation of the construct as an ability, rather than as a set of specific processes and routines. This means that only those items were included which avoided asking "how extensively do you apply process X (a *process* that *aims* at knowledge absorption)?" but, instead, asked "how successful are you in terms of Y (an *aspect* of knowledge absorption)?"

Formal and informal integration mechanisms can be implemented at both types of interface, thus, requiring four variables: formal and informal intra-functional integration mechanisms, as well as formal and informal cross-functional integration mechanisms. Informal integration was measured by means of four items from previous studies (Zahra and Nielsen, 2002) as a reflective scale, thus, indicating the degree of a latent informal integration. While functional integration has been measured unidimensional in the past, for example, in terms of extensive use of cross-disciplinary teams within the R&D function (Henderson and Clark, 1990), formal integration mechanisms have been adopted from previous literature in terms of a formative, multi-dimensional scale (Gupta and Govindarajan, 2000; Jansen *et al.*, 2005). The formative scale of formal integration mechanisms was also measured by means of an additional item in order to achieve a more complete construct, which is particularly important for formative constructs (Edwards and Bagozzi, 2000).

Our model also includes several control variables. We analyse *industry effects* by means of the variable IND2ROS, that is the industry average return on sales (ROS) calculated based on the two-digit ATECO code (Coombs and Bierly, 2006). Alternatively, common industry dummies based on two-digit ATECO codes have been used in OLS regression analyses as a further robustness check (Cassiman and Veugelers, 2006). We control for *firm size* by applying one of the most common measures: the logarithm of the number of full-time equivalent employees. We control for the *degree of centralisation* applying the measure used by Jaworski and Kohli (1993) and for the degree to which the firm or business unit *directly serves end consumers* (B2C) rather than other businesses (B2B) using the scale applied by (Verhoef and Leeftang, 2009). *Environmental turbulence* is controlled for by means of a formative item battery (Verhoef and Leeftang, 2009). Finally, we control for *market oriented reward mechanisms* using items from Jaworski and Kohli (1993).

Data

The sample in the present study includes responses from 126 professionals in R&D and/or M&S departments from 79 different firms (in 37 cases, we received a response from only one department). Although it has been found that Structural Equation Models (SEM) can perform well, even with sample sizes as low as 50 (Iacobucci, 2010), it is well known that one should aim for sample sizes above 100 (preferably 200) cases (Bagozzi and Yi, 2012). While our sample size is admittedly towards the lower bound of admissible sample sizes, other recent studies have applied SEM on similar topics with similar values (Stock *et al.*, 2014, for example, use a sample size of 125). However, in order to further check the impact of sample size on the robustness of our models in terms of stable parameter estimates, we tested our model initially by means of the first 100 cases and, subsequently, by means of the final set of 126 cases. There was no indication of bias.

Missing data can have serious effects on data quality and the conclusions that can be drawn from empirical data (Byrne, 2010; Schafer and Graham, 2002). The variables with most missing cases showed less than 10% of cases missing, which would still sufficiently limit potential bias (Schafer and Graham, 2002). The two most common imputation techniques have been chosen for the missing data: variable means and maximum likelihood estimation as available in AMOS (cf. Byrne, 2010). Moreover, all analyses of the measurement and structural models were performed based on both types of imputation techniques showing no substantial differences. Since the maximum likelihood estimation of missing observations is generally regarded as robust even in cases where the missing-at-random condition is not exactly met (Byrne, 2010), we report results based on this imputation technique.

Common method bias was checked by means of Harman's single-factor test that is commonly applied in cross-sectional studies (Harman, 1967). Thus, a principal component analysis of all items of the survey extracting factors with eigenvalues above 1 has been performed, which resulted in many factors explaining about 75% of total variance and another principal component analysis constraining the extraction of one single factor of the unrotated solution. This single factor could explain only about 23.8% of total variance. Thus, we found no indication that common method bias is a major problem. To reduce the potential of social desirability bias, particular care was taken to indicate full anonymity of both individuals and firms. In order to avoid biases due to the order of items, the items of all multi-item scales were presented in random order each time the site was accessed.

Model specification

Given the low sample size, we tested the measurement and the structural model separately (Anderson and Gerbing, 1988). First, the pure measurement model of the reflective scales was estimated in a confirmatory factor analysis. In this way, we estimated not only composite reliability (CR), but also convergent and discriminant validities. We subsequently used the derived composite variables for the estimation of the pure structural model in a path analysis. An important criterion is the logic of causal direction theorised, which should be confirmed by high inter-item correlations in case of reflective scales (Edwards and Bagozzi, 2000) since, if there is a common latent factor influencing the items, these will have to be correlated to some degree. In contrast, there is no such constraint if the indicators *form* the latent variable. Indeed, for all scales for which we argue that are formative, the inter-item correlations, although significant, are not as high as one might expect if they were reflective. While commonly formative scales are simple, that is, non-weighted averages of the equally scaled items, formal integration mechanisms have been differently summed in extant literature. The items we use to measure cross-functional integration or interfaces have been combined into a weighted average in previous studies, the weights being 1 for liaison personnel, 2 for temporary task forces, and 3 for permanent teams (Gupta and Govindarajan, 2000; Jansen *et al.*, 2005). The same weights have been applied in this study, while the additional indicator “job rotation” is a weighted one, because by nature it is closest to liaison personnel as it involves only single individuals.

Results

Measurement model results

Items of reflective scales have been found to correlate at least over 0.45 and significantly so, mostly at the 1%-level. This is in line with the pattern of factor loadings identified by the exploratory factor analysis where all items of the reflective scales load together on their respective factors, with the exception of a few items of the FAC and CFAC scales. The confirmatory factor analysis led to the elimination of several items from the original scales because factor loadings were too low (< 0.55). Moreover, a test for metric invariance during confirmatory factor analysis revealed that FAC and CFAC have to be measured by different items. This further corroborates that the two concepts are actually different. Both the Tucker–Lewis-Index (TLI) and the comparative fit index (CFI) are over 0.9 with values of 0.921 and 0.941, respectively, the root mean square error of approximation (RMSEA) at 0.063 being in a well acceptable range (< 0.1 moderate;

< 0.05 good), and chi-square/d.f. at 1.491 being below the conservative threshold of 2. Model fit is at least as good for the solution with variable means imputed data, for which the estimate of SRMR was well below the threshold of 0.08, too (Hu and Bentler, 1999). Moreover, no indication of problems with model fit could be found following a check of the matrix of the standardised residual covariances available for the analysis with mean imputed variable means, since no value is larger than 2.58 (cf. Byrne, 1999, p. 86). The values for CR of all scales were largely above the 0.7 threshold (Bagozzi and Yi, 1988). Moreover, for all variables, convergent and discriminant validity is supported, since average variance extracted (AVE) is always smaller than CR and larger than both maximum shared squared variance (MSV) and average shared squared variance (ASV).

Structural model results

The structural model is based on the specified hypotheses and includes the specified control variables. Table 1 and Fig. 1 summarise our results. In multi-group SEM models, an R-square value is estimated contemporaneously in one model for every endogenous variable for each group. All six of our R-square values are between 0.35 and 0.59 and hence show that our model explains a relatively large percentage of variance of the endogenous variables.

Our first hypotheses could not be confirmed. We found no significant effects of either informal (H1a) or formal (H1b) intra-functional integration mechanisms on FAC. A possible explanation is that, within functional areas, vertical information flows are more important than lateral ones and complementarities are realised directly and exclusively in projects rather than at the level of departments.

Another possible explanation is that FAC depends more on in-depth knowledge since it concerns the absorption of knowledge from the same domain (e.g., a marketing department in the USA absorbing knowledge from a marketing department in Europe). In this case, successful absorption hinges more on the departments' knowledge bases being already very similar rather than their organisational integration. By contrast, where knowledge bases are always relatively distant and little common ground exists, integration mechanisms and higher frequency of exchange substitute for the common ground provided by a similar domain background. In fact, the positive effect of informal integration across functional domains on FAC is found to be highly significant for both R&D and M&S departments, which confirms our Hypothesis 2.

Likewise, we find support for Hypotheses 3a and 3b. We can confirm for R&D departments that formal cross-functional integration mechanisms highly significantly and positively affect CFAC, while informal cross-functional integration mechanisms have a slightly significant negative effect on CFAC. Moreover, we

Table 1. Structural path model results with control variables (ML estimation of missing data).

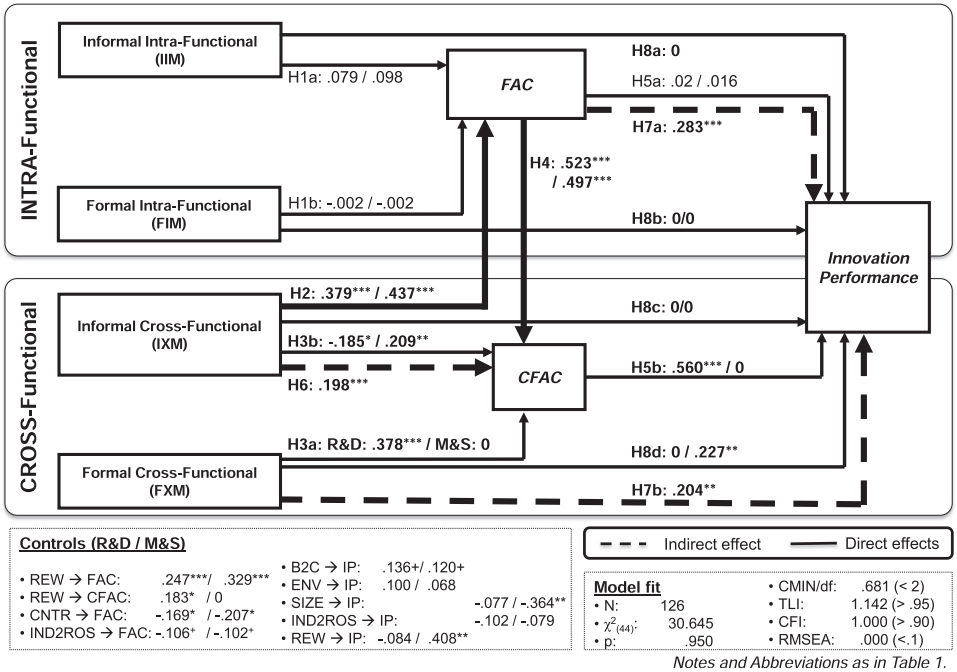
DV	Hypothesis	Path	R&D		M&S		Expected	Confirmed		
			R ²	β	(p-val.)	R ²			β	(p-val.)
FAC			0.357			0.590				
	H1a:	FAC ← IIM		0.079	(0.362)		-0.098	(0.362)	+	No
	H1b:	FAC ← FIM		-0.002	(0.982)		-0.002	(0.982)	+	No
	H2:	FAC ← IXM		-0.379***	(0.000)		-0.437***	(0.000)	+	Yes
	C1:	FAC ← CNTR		-0.169*	(0.013)		-0.207*	(0.013)	-	Yes
	C2:	FAC ← IND2ROS		-0.106 ⁺	(0.112)		-0.102 ⁺	(0.112)	-	~
CFAC	C3:	FAC ← REW		-0.247***	(0.000)		-0.329***	(0.000)	+	Yes
			0.529			0.407				
	H3a (R&D):	CFAC ← FXM		-0.378***	(0.000)				+	Yes
	H3a (M&S):	CFAC ← FXM					0	(n.a.)	0	Yes
	H3b (R&D):	CFAC ← IXM		-0.185 ⁺	(0.108)				-	~
	H3b (M&S):	CFAC ← IXM					-0.209 ⁺⁺	(0.073)	+	~
IP	H4:	CFAC ← FAC		-0.523***	(0.000)		-0.497***	(0.000)	+	Yes
	C4 (R&D):	CFAC ← REW		-0.183*	(0.046)				+	Yes
	C4 (M&S):	CFAC ← REW					0	(n.a.)	0	Yes
			0.372			0.356				
	H5a:	IP ← FAC		-0.020	(0.847)		-0.016	(0.847)	+	No
	H5b (R&D):	IP ← CFAC		-0.560***	(0.000)				++	Yes
H5b (M&S):		IP ← CFAC					0	(n.a.)	0	Yes
	H8a:	IP ← IIM		0	(n.a.)		0	(n.a.)	0	~
	H8b:	IP ← IXM		0	(n.a.)		0	(n.a.)	0	~
	H8c:	IP ← FIM		0	(n.a.)		0	(n.a.)	0	~
	H8d (R&D):	IP ← FXM		0	(n.a.)				0	~
	H8d (M&S):	IP ← FXM					0.227 ⁺⁺	(0.061)	+	~

Table 1. (Continued)

DV	Hypothesis	Path	R&D		M&S		Expected	Confirmed	
			R ²	β	(p-val.)	R ²			β
	C5:	IP ← B2C		−0.136 ⁺	(0.101)	−0.120 ⁺	(0.101)	+	~
	C6:	IP ← ENV		−0.100	(0.255)	−0.068	(0.255)	+	No
	C7 (R&D):	IP ← SIZE		−0.077	(0.425)			−	No
	C7 (M&S):	IP ← SIZE				−0.364**	(0.002)	−	Yes
	C8:	IP ← IND2ROS		−0.102	(0.204)	−0.079	(0.204)	−	No
	C9 (R&D):	IP ← REW		−0.084	(0.487)			+	No
	C9 (M&S):	IP ← REW				−0.408**	(0.002)	+	Yes
Indirect effects of integration mechanisms									
	H6:	CFAC ← FAC ← IXM		−0.198***	(0.000)			+	Yes
	H7a:	IP ← CFAC ← FAC		−0.283***	(0.000)			+	Yes
	H7b:	IP ← CFAC ← FXM		−0.204**	(0.000)			+	Yes

Notes: $n = 126$; $+ < 0.11$, $++ < 0.07$, $* < 0.05$, $** < 0.01$, $*** < 0.001$; $\chi^2 = 30.645$; d.f. = 45; $p = 0.950$; χ^2 /d.f. = 0.681; TLI = 1.142; CFI = 1.00; RMSEA = 0.000 (90% confidence interval 0.000 $\rightarrow$ 0.003); SRMR = not defined for data with missing values. β = standardised parameter estimates. Significance levels of indirect effects based on two-tailed Sobel-test.

Hypotheses regarding zero effects have been tested by constraining the parameters in question to zero and calculate the chi-square difference test statistic to compare it with the unconstrained model. Acceptance based on chi-square difference test always coincided with decision if based on AIC. **Abbreviations:** Innovation Performance (IP), Functional AC (FAC), Cross-functional AC (CFAC), Informal (IXM) and formal (FXM) cross-functional integration mechanisms, Informal (IIM) and formal (FIM) intra-functional integration mechanisms, Market oriented reward mechanisms (REW), Centralisation (CNTR), Environmental turbulence (ENV), Industry average of Return on Sales (IND2ROS), Percentage of Business-to-Consumer sales of total sales (B2C).



Notes: Abbreviations as in Table 1.

Fig. 1. Overview of structural model results.

find evidence that these effects of formal and informal cross-functional integration mechanisms are, as predicted, partly inverted when considering M&S departments. Constraining the path of formal cross-functional integration mechanisms on CFAC to zero for M&S departments significantly improves model fit, and the effect of informal mechanisms has a positive rather than negative effect that is significant at 7%.

We find support for Hypothesis 4, that is, for the interrelation between FAC and CFAC. FAC has a significant and highly positive impact on CFAC. FAC can thus be regarded as a fundamental capability on which CFAC can build. One may also interpret FAC as an auxiliary capability for successful cross-functional integration. Cross-functional integration processes sometimes require additional information from peer departments, thus making FAC to a certain degree a precondition for CFAC.

We find no support for our Hypothesis 5a, whereby FAC positively affects innovation performance. A possible explanation for the insignificant result regarding Hypothesis 6a is that in most organisations the departmental structure, in particular that of R&D departments, mirrors more or less closely the modular structure of the underlying technology. That means that each of the loosely

coupled modules corresponds to an organisational unit to develop it, and that these units are coupled as loosely as the modules (MacCormack *et al.*, 2012). If this is the case, there are fewer interdependencies between peer departments, between, for example an R&D department responsible for innovations and improvements around module A, and another department responsible for module B. The relevance of knowledge absorption depends on these interdependencies; hence, in these cases, FAC should exhibit low to no impact on innovation performance. In contrast to 5a, we can confirm our Hypothesis 5b: in case of R&D departments, CFAC significantly and positively affects innovation performance, as expected. This effect is robust across diverse model specifications.

We find support for Hypothesis 6: there is a significant, positive indirect effect from cross-functional integration mechanisms through FAC on CFAC. In the case of R&D departments, this indirect effect compensates for the negative direct effect of these mechanisms on CFAC, resulting in a total effect close to zero.

The hypothesised indirect effects from FAC (H7a) and formal cross-functional integration mechanisms (H7b) through CFAC on innovation performance are highly significant, too. The significance levels of all indirect effects are based on a two-tailed Sobel-test. Together with the hypothesis of zero effects for direct paths of cross-functional integration mechanisms on innovation performance, as confirmed above, this might explain previous contrasting results that find negative, positive or no effects at all from these mechanisms on innovation performance. In particular, the indirect positive effect of informal cross-functional integration mechanisms via FAC on CFAC of R&D departments does balance out the negative effect that these mechanisms exert directly on CFAC. This could explain why previous studies did not find any significant effect for informal cross-functional integration mechanisms on integration success (Leenders and Wierenga, 2002).

None of the integration mechanisms included in hypotheses 8a-c affects innovation performance directly. These, and all other zero-effect hypotheses, were tested by means of chi-square difference tests of the nested models and can thus be regarded as not rejected. All models specifying a direct effect of integration mechanisms on innovation performance had to be rejected due to bad model fit. Hence, this confirms our Hypotheses 8a-c. We also find that Hypothesis 8d is confirmed; hence, formal cross-functional integration mechanisms have the expected significant, positive direct effect on innovation performance for M&S departments, whereas they do not show such effect in the case of R&D departments, where the direct effect is zero but the indirect effect is highly significant and positive. This can be interpreted as evidence of the direction of knowledge flow between departments and indirectly supports our assumption that M&S departments do not need CFAC to increase innovation performance, since they deliver

the required knowledge without the necessity to absorb R&D knowledge in turn. In contrast, R&D departments can use all formal integration mechanisms according to their needs; however, without learning how to use these mechanisms to foster knowledge absorption, there will be no significantly positive effect on innovation performance.

Most control variables load as anticipated. Industry sectors, as measured by average return on sales of the sector, have no effect, as confirmed by mostly insignificant industry dummies in OLS regressions (see section on further robustness checks below).

Further robustness checks

We discuss the outcomes of several competing models that have been tested and compared with the base model. In model (1), we assume that FAC is not a prerequisite causing CFAC as argued above, but rather that both variables independently cause innovation performance. Given the differences of the alternative models, we compare them based on Akaike's Information Criterion (AIC). With an AIC value of 391.685, this model has a fit which is worse compared with the base model, which has an AIC value of 356.645.

In model (2), we assume that not only informal but also formal cross-functional integration mechanisms affect FAC. Knowledge about the formal integration mechanisms used across domain boundaries could also be transferred to improve the implementation of these integration mechanisms with other departments of the own corporate function, thus improving FAC. Not only is the newly introduced effect of formal cross-functional integration mechanisms on FAC itself insignificant (p -level 0.253), but the chi-square difference test also fails to indicate a significant difference (p -level 0.269). Moreover, the AIC of this alternative model is higher than that of the original model. Our original model thus overall outperforms this alternative specification as well.

In model (3), we also tested for reversed causality. In this model, we try to address potential endogeneity as much as possible. We assume in this model that higher innovation performance increases managers' perception of their department's ability to absorb new external knowledge. While model 3 performs better than model 1, it performs worse than the original model, based not only on the comparison of the AIC (360.210) but also on all other model fit indicators, except for CFI, which is equally maximised.

Finally, in model (4), OLS regression was applied to double check the principal hypotheses. This allows us to check for robustness not only in terms of model configuration, but also in terms of an alternative and still more common way to assess industry effects, namely by means of the usual industry dummies. All other

controls and composite variables are those used in the structural model. At the same time, OLS also permits a check for multicollinearity issues by means of inspection of the variance inflation factors (VIF). Here, we found all factors ranging between 1 and 3, far below the upper threshold value of 5 (De Vaus, 2002; Kutner *et al.*, 2005). Visual inspection as to heteroskedasticity does not suggest such an issue for any of the OLS models we ran. Generally, the OLS models confirm our principal hypotheses that are also confirmed by the SEM model. Moreover, we cannot exclude firm-specific effects, but since we had only very few cases per firm, we can relatively safely assume unrelated residual errors. However, as a robustness check, we created a dummy variable for those firms from which we had received only one response. It is not possible to include dummies in an ML estimation-based SEM, but we controlled for the influence of this dummy on the main results in the OLS regressions. Our findings were still supported when controlling for number of responses per firm.

Discussion and Conclusion

We argued that a crucial aspect of integration success at inter-departmental boundaries in terms of innovation performance is the capability of the organisational units to absorb each other's knowledge. We hence empirically studied integration within and between the M&S and R&D functions and were able to explain a considerable share of the variance of innovation performance. Previous research on this phenomenon investigated the direct relationship between differently operationalised integration mechanisms and innovation performance without taking into account the nature of the units' corporate function. This led to contrasting empirical results and explanations. Hence, we investigated whether the ability of departments to absorb knowledge from other departments can improve innovation performance, as well as whether such an ability can be built by means of integration mechanisms.

Based on our analysis, however, we found evidence of contrasting direct and indirect effects of some integration mechanisms on CFAC and hence innovation performance. In fact, it is important to note that, in case of R&D departments, the use of informal cross-functional integration mechanisms has a positive indirect effect on CFAC mediated by FAC, while, at the same time, it has a negative direct effect on CFAC. That is, it generally improves the ability to understand what types of external knowledge from within the department's own functional domain are most valuable given the complementarity with other functional domains, but it may likewise provide too much potentially contradictory information that hinders the integration of these other functional domains. Since the indirect effect of

informal cross-functional integration via FAC on CFAC is highly significant and positive, while the direct effect is significantly negative, the total effect on CFAC — and thus innovation performance — is close to zero. This is in line with previous findings that did not consider the mediation but only the aggregate effect, and hence did not find a significant relation (Leenders and Wierenga, 2002).

Moreover, the CFAC of R&D departments enables them to improve innovation performance. We found this construct, which we conceptualise for the first time at this level of analysis, to be an important mediating variable. It shows that the mere implementation of integration mechanisms is not a sufficient criterion for improved innovation performance. R&D departments have to learn how to use them and how to absorb knowledge from M&S in order to develop more valuable and better performing innovations. We also show that M&S departments need not be particularly skilled in absorbing knowledge from R&D departments to provide the necessary updated market knowledge to R&D departments. While these findings confirm previous studies on the direction of knowledge flow from M&S to R&D departments in new product development, they also contribute to the clarification of previous contrasting findings regarding the effects of integration mechanisms on innovation performance.

Several limitations have to be mentioned, however. First, the results are limited so far to the Italian context, and a cross-national replication would add to the reliability of the generalisation of the results. Secondly, the fact that the study is based on cross-sectional data makes the causal directions depend fundamentally on the theory developed. A future study based on panel data could add to the strength of the causal inference, as changes over time in terms of departmental FAC and CFAC and their impact on innovation performance could actually then be analysed. This model can be refined using scales that better reflect the theorised multi-dimensionality. However, these simpler preliminary scales could be used already for further inquiries. A third limitation is the relatively small sample size. The statistical power of the model is relatively limited; in particular the hypotheses of zero effects have to be regarded with caution, since, with a comparatively low power, the probability of false negatives increases.

Theoretical implications

We conceptualise Functional and Cross-functional Absorptive Capacity (FAC and CFAC, respectively) and apply technically valid and reliable empirical measurement instruments for these concepts. In doing so, we follow similar considerations expressed in the conceptualisation of relative AC at the firm level (Lane and Lubatkin, 1998) and address highly relevant yet under-researched issues of the AC literature (Volberda *et al.*, 2010). Hence, this study contributes to

theory in several important ways. We provide theoretical underpinning for the relevance of AC (in form of FAC and CFAC) for functional integration and innovation performance. Based on an original database from Italian manufacturing firms from a broad range of different sectors, we also provide the first empirical evidence for the role that departmental AC plays in functional integration. Hence, our study also provides quantitative evidence of CFAC's positive mediation of the effect that formal cross-functional integration mechanisms have on innovation performance.

Furthermore, we contribute to the theory of cross-functional integration through analysing the role of AC in this context, thus providing a deeper understanding of the integration process. Our differentiated view, based on the revised conceptualisation of AC helps to explain previously contrasting results in the literature (Troy *et al.*, 2008). Departments' absorptive capacities are specific to the type of intra-organisational boundary; hence, these distinct capabilities mediate differently the effects of different integration mechanisms (i.e., formal and informal) on innovation performance. In particular, with our distinction of two fundamental types of integration mechanisms (formal versus informal) at two different types of organisational boundaries (intra-versus cross-functional), we allow for a nuanced understanding of how departmental ACs affect innovation performance.

Finally, including and comparing R&D and M&S departments in one joint analysis, we can show how these effects differ across different corporate functions. In this way, we contribute to a better understanding of the differences in successful functional integration and to an explanation for the contrasting findings on the role of integration for innovation performance (Troy *et al.*, 2008).

Clearly, there are many new research questions implied by our findings, of which we can only point to some we view as most relevant. Regarding functional and cross-functional integration, the recognition of these mediating effects permits more fine-grained analyses of contingency theories and moderating effects regarding differentiation and reintegration in functionally specialised organisations. Future research hence could, for example, address the issue of intra-firm heterogeneity regarding not only the development, but also the use of these abilities. Furthermore, future research could analyse what this means in the context of multinational corporations and globally dispersed innovation activities as well as in terms of recent contingency views on integration (Burgers and Covin, 2016). Multinational corporations have to differentiate and reintegrate in different environments and different degrees of innovation dynamism. On one hand, departmental ACs could help to explain heterogeneity on the levels of success of integration across countries. A multinational that tries to integrate knowledge inflows from its foreign subsidiaries should perform better when its receiving home-country departments exhibit high levels of AC. On the other hand, the

multinational context could help to explain differences in the way and speed in which geographically dispersed departments develop these ACs. For example, R&D departments of a multinational corporation are necessarily integrated more through formal integration mechanisms, since geographical distance reduces the likelihood of informal exchange, leading thus to higher CFAC.

Regarding AC, this concept is still mostly theorised and observed at the firm-level. While most recent research tries to connect individual employee-level directly to firm-level AC (Distel, 2018; Ebers and Maurer, 2014; Yao and Chang, 2017) in order to establish micro-foundations (Felin *et al.*, 2012) more research is needed to develop sound micro-foundations for AC on the department-level as well, for which we offer a starting point. This means appreciating the fact that, in hierarchical organisations with many levels, the observation and analysis of meso-level phenomena helps one to understand the emergence of more macro-level phenomena.

Managerial implications

Our findings also have several implications for managing departments involved in new product development. In particular, we demonstrate that informal, spontaneous communication might pose serious pitfalls for R&D departments and for the new product development process altogether. This is in line with and adds to rather recent findings (Sosa *et al.*, 2015). We find a positive effect on FAC and a negative one on CFAC. This could be leveraged by focusing on strengthening the positive first effect and avoiding the second negative one. First, firms should focus on the scope of informal communication between M&S and R&D departments on the product development stages in which such communication helps R&D departments understand the markets and, hence, to develop requirements and targets of their peer R&D departments with which they cooperate. The informal communication with M&S could thus complement and integrate the rather technical communication with their R&D peers. On the other hand, to avoid any negative effect on CFAC, continuous interruptions of their own development process through informal feedback at arbitrary development stages should be avoided, because they disrupt the workflow. This is in line with recent findings based on social network analysis in complex new product development (Sosa *et al.*, 2015). Companies can benchmark their departments based on our scales in order to judge the need to pay more attention to the development of departmental absorptive capacities, thereby improving strategic decisions in innovation management. This allows firms to allocate the appropriate level of attention, time, et cetera to build departmental absorptive capacities.

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Appendix

Table A1. Core scales used in the questionnaire.

			Mean (SD)	
Cross-/Functional AC (CFAC/FAC)	(reflective)	Item based on:	FAC	CFAC
Members of our department. . .				
(5) quickly analyse and interpret changing opportunities of “technologies”/“markets”.		Jansen <i>et al.</i> (2005)	4.17 (1.55)	4.10 (1.44)
(6) structure and integrate new external knowledge with ease.		Jansen <i>et al.</i> (2005)	a	3.94 (1.46)
(9) recognise timely the consequences of new external knowledge to our mode of operation.		(Flatten <i>et al.</i> 2011)	4.37 (1.51)	a
(10) are able to apply new external knowledge in their practical work.		Flatten <i>et al.</i> (2011)	4.59 (1.51)	a
(11) regularly reconsider their knowledge and adapt it according to new external knowledge.		Jansen <i>et al.</i> (2005)	4.52 (1.53)	4.13 (1.46)
(12) know to share and apply new external knowledge with those in our department who is most apt.		Szulanski (1996)	4.19 (1.53)	a
Informal Integration (IIM & IXM Scales)			Mean (SD)	
Based on Zahra and Nielsen (2002).	(reflective)		IIM	IXM
To what extent does your department use . . .				
(1) free exchange of operating and financial information, with . . .			a	3.06 (1.80)
(2) bypassing of formal communication channels, as needed, with . . .			a	3.53 (1.90)
(3) informal relationships for getting things done, with . . .			4.71 (1.69)	4.05 (1.84)
(4) maintains open communication channels in its operations with . . .			4.61 (1.63)	a
other departments of the (own/other function ^a)				

Table A.1. (Continued)

Formal Integration (FIM & FXM Scales)		Item weights		Mean (SD)	
Based on Gupta and Govindarajan (2000), Jansen <i>et al.</i> (2005).		Extant	Adj.	FIM	FXM
To what extent does your department use:					
(1) liaison personnel with ...		1	1	3.94 (1.77)	3.52 (1.98)
(2) temporary task forces with ...		2	2	4.01 (1.86)	3.28 (1.94)
(3) permanent teams with ...		3	3	4.32 (1.86)	3.66 (2.03)
(4) job rotation with ...		N.A.	1	2.98 (1.88)	2.02 (1.37)
other departments of the (own/other function ^b)					
Innovation Performance (IP)					
Based on Atuahene-Gima <i>et al.</i> (2005)		(reflective)		Mean (SD)	
Rate how your business unit is performing on the following new product development objectives relative to your firm's stated objectives:					
(1) Market share				4.37 (1.25)	
(3) ROI				4.35 (1.37)	
(4) Profitability				4.47 (1.42)	

Notes: Scaling for CFAC, FAC, IIM, IXM, FIM, FXM: 1 — “No or very little extent” ... 7 — “Very large extent”

Scaling for IP: 1 — “much worse” ... 7 — “much better”.

^aItem deleted.

^b(own function) replaced with “R&D function” for R&D departments and “M&S function” for M&S departments (FIM); (other function) replaced with “M&S function” for R&D departments and “R&D function” for M&S departments (FXM).

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